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Mathematics-IV (MTH-241)

Assignment-IV

Duality

Session: 2024-25

1. XYZ Company has three departments – Assembly, Painting and Packing. The company can make three types of almirahs. An almirah of type I requires one hour of assembly, 40 minutes of painting and 20 minutes of packing time, respectively. Similarly, an almirah of type II needs 80 minutes, 20 minutes and one hour, respectively. The almirah of type III requires 40 minutes each of assembly, painting and packing time. The total time available at assembly, painting and packing departments is 600 hours, 400 hours and 800 hours, respectively.

Determine the number of each type of almirahs that should be produced in order to maximize the profit. The unit profit for types I, II and III is Rs 40, 80 and 60, respectively. Suppose that the manager of this XYZ company is thinking of renting the production capacities of the three departments to another almirah manufacturer – ABC company. ABC company is interested in minimizing the rental charges. On the other hand, the XYZ company would like to know the worth of production hours to them, in each of the departments, in order to determine the rental rates.

- (a) Formulate this problem as an LP problem and solve it to determine the number of each type of almirahs that should be produced by the XYZ company in order to maximize its profit.
- (b) Formulate the dual of the primal LP problem and interpret your results.
2. Write the dual of the following primal LPP.

$$\text{Maximize } Z = 3x_1 + x_2 + 2x_3 - x_4$$

Subject to the constraints

$$2x_1 - x_2 + 3x_3 + x_4 = 1$$

$$x_1 + x_2 - x_3 + x_4 = 3$$

and $x_1, x_2 \geq 0$ and x_3, x_4 unrestricted in sign.

3. A company wants to produce three products: A, B and C. The per unit profit on these products are Rs 4, Rs 6 and Rs 2, respectively. These products require two types of resources, manpower and raw material. The LP model formulated for determining the optimal product mix is as follows:

$$\text{Maximize } Z = 4x_1 + 6x_2 + 2x_3$$

subject to the constraints:

$$x_1 + x_2 + x_3 \leq 3, \text{ (Manpower required)}$$

$$x_1 + 4x_2 + 7x_3 \leq 9, \text{ (Raw material available)}$$

and

$$x_1, x_2, x_3 \geq 0,$$

where x_1, x_2, x_3 represent the number of units of products A, B, and C, respectively, to be produced.

- (a) Find the optimal product mix and the corresponding profit of the company.
- (b) Find the range of the profit contribution of product C (i.e., coefficient c_3 of variable x_3) in the objective function, such that the current optimal product mix remains unchanged.
- (c) What shall be the new optimal product mix when per unit profit from product C is increased from Rs 2 to Rs 10?
- (d) Find the range of the profit contribution of product A (i.e. coefficient c_1 of variable x_1) in the objective function such that the current optimal product mix remains unchanged.

4. Solve the following LP problem:

$$\text{Maximize } Z = 4x_1 + 6x_2 + 2x_3$$

subject to the constraints:

$$x_1 + x_2 + x_3 \leq 3,$$

$$x_1 + 4x_2 + 7x_3 \leq 9,$$

and

$$x_1, x_2, x_3 \geq 0.$$

- (a) Discuss the effect of discrete change in the availability of resources from $[3, 9]^T$ to $[9, 6]^T$.
- (b) Which resource should be increased (or decreased) in order to get the best marginal increase in the value of the objective function?

5. Solve the following LP problem:

$$\text{Maximize } Z = -x_1 + 3x_2 - 2x_3$$

subject to the constraints:

$$3x_1 - x_2 + 2x_3 \leq 7,$$

$$-2x_1 + 4x_2 \leq 12,$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10,$$

$$x_1, x_2, x_3 \geq 0.$$

Discuss the effect of the following changes in the optimal solution:

- (a) Determine the range for discrete changes in the coefficients a_{13} and a_{23} consistent with the optimal solution of the given LP problem.
- (b) The x_1 -column in the LP problem is changed from $[3, -2, -4]^T$ to $[3, 2, -4]^T$.
- (c) The x_3 -column in the LP problem is changed from $[2, 0, 8]^T$ to $[3, 1, 6]^T$.

6. Consider the following LP problem:

$$\text{Maximize } Z = 3x_1 + 5x_2$$

subject to the constraints:

$$3x_1 + 2x_2 \leq 18,$$

$$x_1 + 2x_2 \leq 4,$$

$$x_2 \leq 6,$$

$$x_1, x_2 \geq 0.$$

Obtain an optimal solution of the given LP problem.

- (a) Suppose variable x_6 is added to the given LP problem. Then obtain an optimal solution to the resulting LP problem. It is given that the coefficients of x_6 in the constraint of the problem are 1, 1, and 1, and that its coefficient in the objective function is 2.
 - (b) Discuss the effect on the optimal basic feasible solution by adding a new constraint $2x_1 + x_2 \leq 8$ to the given set of constraints.
7. The owner of a readymade garments store sells two types of shirts – Zee-shirts and Button-down shirts. He makes a profit of Rs 3 and Rs 12 per shirt on Zee-shirts and Button-down shirts, respectively. He has two tailors, A and B, at his disposal, for stitching the shirts. Tailors A and B can devote at the most 7 hours and 15 hours per day, respectively. Both these shirts are to be stitched by both the tailors. Tailors A and B spend 2 hours and 5 hours, respectively, in stitching one Zee-shirt, and 4 hours and 3 hours, respectively, on stitching a Button-down shirt. How many shirts of both types should be stitched in order to maximize the daily profit?
- (a) Formulate and solve this problem as an LP problem.
 - (b) If the optimal solution is not integer-valued, use Gomory technique to derive the optimal integer solution.

8. Solve the following mixed-integer programming problem

$$\text{Maximize } Z = x_1 + x_2$$

subject to the constraints

$$3x_1 + 2x_2 \leq 5,$$

$$x_2 \leq 2,$$

$$x_1, x_2 \geq 0, x_1 \text{ is a non-negative integer.}$$

9. Solve the following all-integer programming problem using the branch and bound method.

$$\text{Maximize } Z = 3x_1 + 5x_2$$

subject to the constraints

$$2x_1 + 4x_2 \leq 25,$$

$$x_1 \leq 8,$$

$$2x_2 \leq 10,$$

$$x_1, x_2 \geq 0 \text{ and integers.}$$

10. Is it possible to solve the following linear programming problem with graphical method? Also, solve it using the Cutting Plane Algorithm.

$$\text{Maximize } 2x_1 - 4x_2 + x_3$$

subject to

$$x_1 - x_2 \leq 12,$$

$$2x_2 + 3x_3 \leq 28,$$

$$x_1, x_2, x_3 \geq 0 \quad \text{and integral.}$$